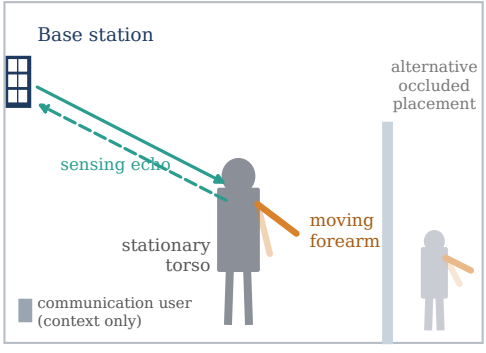
(a) One sensing task



Same room. Same arm motion. Same deadline.

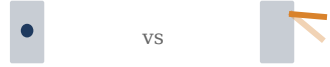
Detect motion onset within 200 ms

3.5 GHz | 12 GHz | 15 GHz | 28 GHz

FR3

(b) Three modeling choices

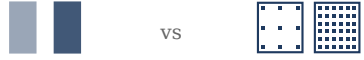
Target



Stationary point: no independent limb motion

Articulated target: motion enters the echo

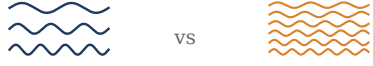
Antenna



Fixed gain (aperture shrinks with f)

Aperture-based gain scaling (idealized comparison)

Detector



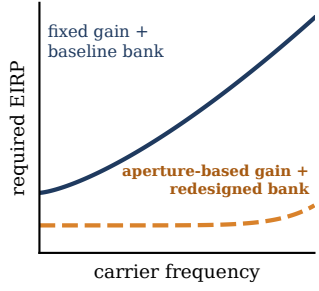
Baseline bank (coarse fixed grid)

Redesigned bank (better-placed templates)

(c) The design question

How much sensing EIRP does each band need?

(conceptual illustration, not data)



Interpret a band comparison together with its target, antenna, and detector assumptions.